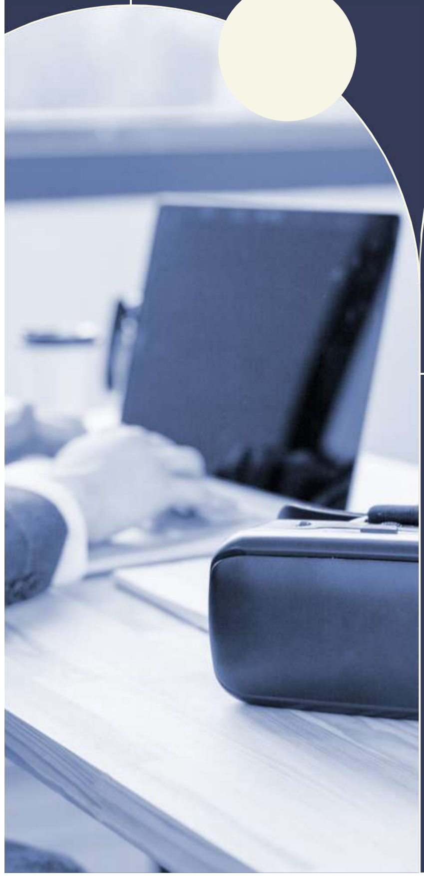


Profile Monitoring
輪廓監控

李俊毅
國立成功大學統計學系

大綱

01 Profile Monitoring

❖ Introduction to Profile Monitoring

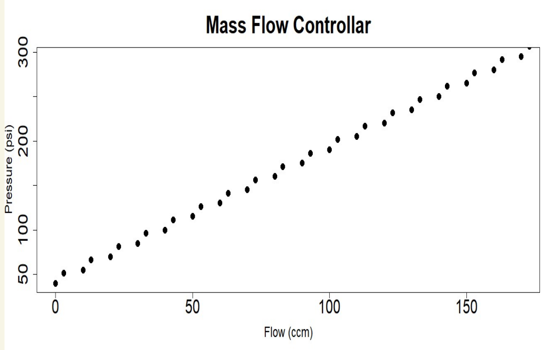
02 ❖ Two Applications

Background

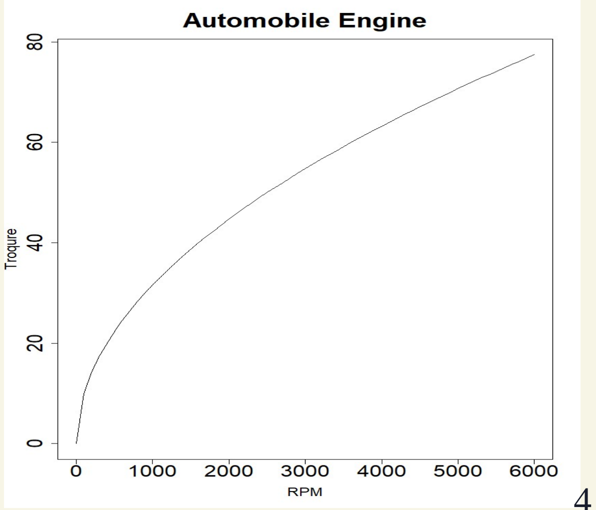
- **SPC is an important method for monitoring product and process quality.**
  - **Traditionally, a single or a vector quality characteristics is used to measure the process or product quality at a given time or space.**
- **However, the quality of a product can be characterized by a function is denoted as profile.**
- **A profile describes a relationship between the response variable and one or more explanatory variable.**

Real Examples

- **Mass flow Controller (Pressure vs flow)**



- **Automobile engine (Torque vs speed)**



Profile Control Chart (Shewhart)

- **Simple Shewhart control chats**
  - **Model**

$$y_{ij} = \beta_0 + \beta_1 \mathrm{x_i} + \varepsilon_{ij}, \mathrm{i} = 1,2,\ldots,\mathrm{n}, \mathrm{j} = 1,2,\ldots \quad \varepsilon_{ij} \sim N(0,\sigma^2)$$

$$= \beta_0 + \beta_1(\mathrm{x_i} - \bar{X}) + \varepsilon_{ij}$$

  - **Monitoring statistics**

$$b_{1j} = \hat{\beta}_{1j} = \frac{\sum_{i=1}^{n}(x_i - \bar{X})(x_i - \bar{y})}{\sum_{i=1}^{n}(x_i - \bar{X})^2} \sim N(\beta_1, \sigma^2 / \sum_{i=1}^{n}(x_i - \bar{X})^2)$$

$$b_{0j} = \hat{\beta}_{0j} = \bar{y} \sim N(\beta_0, \sigma^2/n)$$

$$\hat{\sigma}_j{}^2 = \frac{1}{n}\sum_{i=1}^{n} e_{ij}^2, \quad e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - (\hat{\beta}_{0j} + \hat{\beta}_{1j} x_i)$$

An Example

- **Simple Shewhart control charts**
  - **Probability Model**

$$1-\alpha = \Pr(LCL \le b_{1j} \le UCL \mid \beta_1)$$
$$= \Pr\left(\frac{LCL-\beta_1}{\sqrt{\sigma^2/S_{XX}}} \le Z \le \frac{UCL-\beta_1}{\sqrt{\sigma^2/S_{XX}}}\right)$$

$$1-\alpha = \Pr(LCL \le b_{0j} \le UCL \mid \beta_0)$$
$$= \Pr\left(\frac{LCL-\beta_0}{\sqrt{\sigma^2/n}} \le Z \le \frac{UCL-\beta_0}{\sqrt{\sigma^2/n}}\right)$$

  - **Control Limits**

$$UCL = \beta_1 + Z_{\alpha/2}\sqrt{\frac{\sigma^2}{S_{XX}}}$$
$$CL = \beta_1$$
$$LCL = \beta_1 - Z_{\alpha/2}\sqrt{\frac{\sigma^2}{S_{XX}}}$$

$$UCL = \beta_0 + Z_{\alpha/2}\sqrt{\frac{\sigma^2}{n}}$$
$$CL = \beta_0$$
$$LCL = \beta_0 - Z_{\alpha/2}\sqrt{\frac{\sigma^2}{n}}$$

An Example

- **Simple Shewhart control charts**
  - **Probability Model**

$$
\begin{aligned}
1-\alpha &= \Pr\left(LCL \le \hat{\sigma}_j^2 \le UCL \middle| \sigma^2\right) \\
&= \Pr\left(\frac{(n-2)}{\sigma^2} LCL \le \frac{(n-2)\hat{\sigma}_j^2}{\sigma^2} \le \frac{(n-2)}{\sigma^2} UCL\right)
\end{aligned}
$$

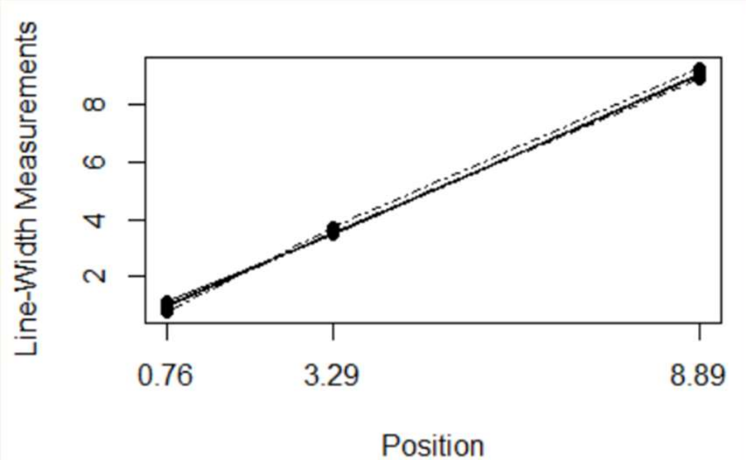
  - **Control Limits**

$$
\begin{aligned}
UCL &= \frac{\sigma^2}{(n-2)} \chi^2_{\alpha/2,(n-2)} \\
CL &= \sigma^2 \\
LCL &= \frac{\sigma^2}{(n-2)} \chi^2_{1-\alpha/2,(n-2)}
\end{aligned}
$$

Numerical Example

Day	Position	x	y
1	L	0.76	1.12
1	M	3.29	3.49
1	U	8.89	9.11
2	L	0.76	0.99
2	M	3.29	3.53
2	U	8.89	8.89
3	L	0.76	1.05
3	M	3.29	3.46
3	U	8.89	9.02
4	L	0.76	0.76
4	M	3.29	3.75
4	U	8.89	9.30
5	L	0.76	0.96
5	M	3.29	3.53
5	U	8.89	9.05
6	L	0.76	1.03
6	M	3.29	3.52
6	U	8.89	9.02

$$y_{ij} = 4.49 + 0.98x_i' + \varepsilon_{ij}\ , i = 1,2,3\ j = 1,2, \dots ,$$
$$x_i' = x_i - \bar{x}\ and\ \varepsilon_{ij} \sim N(0, 0.06826^2)$$



Numerical Example

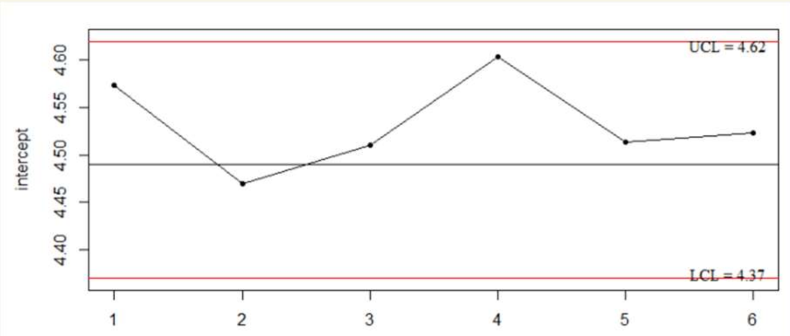
$\alpha' = 0.00167$

$\alpha = 0.005$

$$UCL = \beta_0 + Z_{\alpha'/2}\sqrt{\frac{\sigma^2}{n}} = 4.49 + 3.1434\sqrt{\frac{0.06826^2}{3}} = 4.62$$

$$CL = 4.49$$

$$LCL = \beta_0 - Z_{\alpha'/2}\sqrt{\frac{\sigma^2}{n}} = 4.49 - 3.1434\sqrt{\frac{0.06826^2}{3}} = 4.37$$



$\Pr\{out - of - cotnrol\}$

=$\Pr\{chart1\ OC\ or\ chart2\ OC\}$

=1-$\Pr\{chart1\ IC\ and\ chart2\ IC\}$

=1-$\Pr\{chart1\ IC\}\Pr\{chart2\ IC\}$

=1-$(1 - \Pr\{chart1\ OC\})(1 - \Pr\{chart2\ OC\})$

=1-$(1-\alpha)(1-\alpha)$

=1-$(1-\alpha)^2$

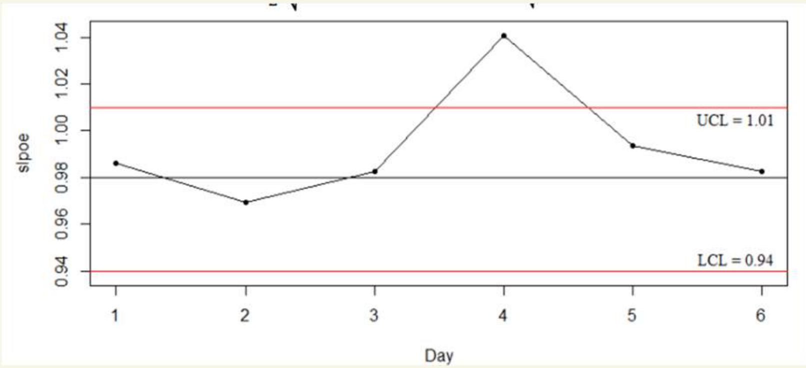
0.005 = $1 - (1 - 0.0167)^3$

Numerical Example

$$UCL = \beta_1 + Z_{\alpha'/2}\sqrt{\frac{\sigma^2}{S_{xx}}} = 0.98 + 3.1434\sqrt{\frac{0.06826^2}{34.62}} = 1.01$$

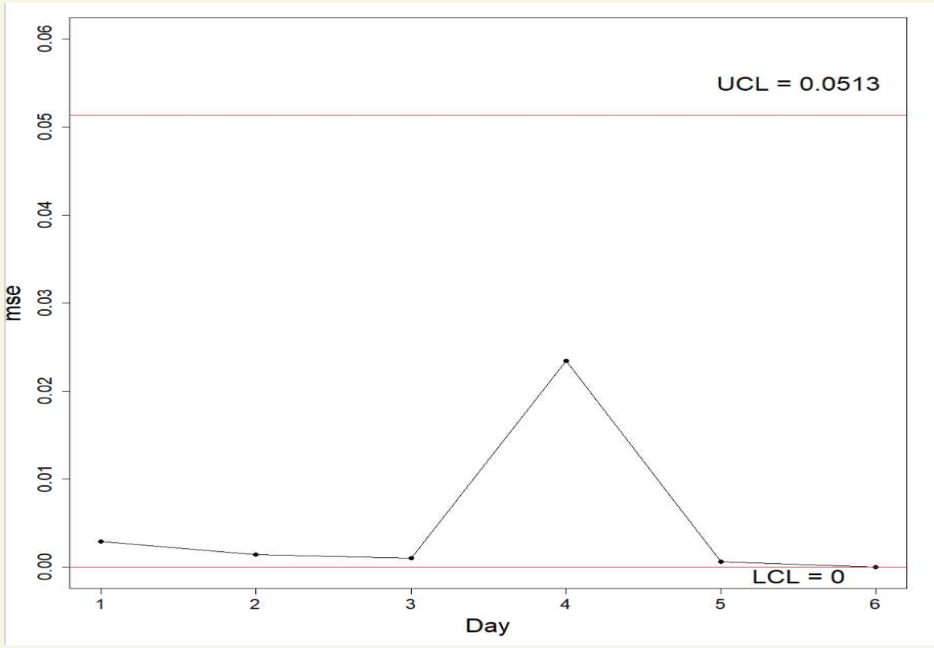
$$CL = \beta_1 = 0.98$$

$$LCL = \beta_1 - Z_{\alpha'/2}\sqrt{\frac{\sigma^2}{S_{xx}}} = 0.98 - 3.1434\sqrt{\frac{0.06826^2}{34.62}} = 0.94$$



Numerical Example

$$UCL = \frac{\sigma^2}{n-2}\chi^2_{\alpha'/2,(n-2)}$$
$$CL = \sigma^2 = 0.0046$$
$$LCL = \frac{\sigma^2}{n-2}\chi^2_{1-\alpha'/2,(n-2)}$$



An Example

- **Multivariate Approach (Kang and Albin, 2000)**
  - **Probability Model**

$$Let\ Z_j = \left(b_{0j}, b_{1j}\right)^T, U = (\beta_0, \beta_1)^T\ is\ the\ expected\ value, \Sigma = \begin{pmatrix} \sigma_0^2 & \sigma_{01}^2 \\ \sigma_{01}^2 & \sigma_1^2 \end{pmatrix}$$

  - **Monitoring statistics**

$$T_j^2 = \left(Z_j - U\right)^T \Sigma^{-1} \left(Z_j - U\right)$$

  - **Upper Control Limits**

$$UCL = \chi_\alpha^2(2)$$

An Example

- **Residual Approach (Kang and Albin, 2000)**
  - **Probability Model**

$$e_{ij} = y_{ij} - \hat{y}_{ij} = y_{ij} - (b_{0j} + b_{1j} x_i)$$

  - **Monitoring statistics**

$$\overline{e_j} = \frac{1}{n}\sum_{i=1}^{n} e_{ij}\,, \overline{e_j} \sim N\left(0, \frac{\sigma}{\sqrt{n}}\right)$$

$$R_j = \max_i(e_{ij}) - \min_i(e_{ij})$$

$$Z_j = \lambda \overline{e_j} + (1-\lambda) Z_{j-1}$$

  - **Upper Control Limits**

$$UCL = L\sigma\sqrt{\frac{\lambda}{n(2-\lambda)}}$$

$$LCL = -L\sigma\sqrt{\frac{\lambda}{n(2-\lambda)}}$$

$$UCL = \sigma(d_2 + Ld_3)$$
$$CL = \bar{R}$$
$$LCL = \sigma(d_2 - Ld_3)$$

Linear Profile

- **MEWMA Approach (Zou et al, 2000)**
  - **Probability Model**

$$y_j = X_j\boldsymbol{\beta} + \varepsilon_j, \varepsilon_j \sim MN(0, \sigma^2 I), j = 1,2, \dots$$

  - **Monitoring statistics**

$$\boldsymbol{Z}_j = \left(Z_j^T(\boldsymbol{\beta}), Z_j(\sigma)\right)^T, Where\ Z_j(\boldsymbol{\beta}) = \frac{\widehat{\boldsymbol{\beta}}_j - \boldsymbol{\beta}}{\sigma}, Z_j(\sigma) = \Phi^{-1}\left[F_{\chi(n-p)}\left(\frac{(n-p)\hat{\sigma}_j^2}{\sigma^2}\right)\right]$$

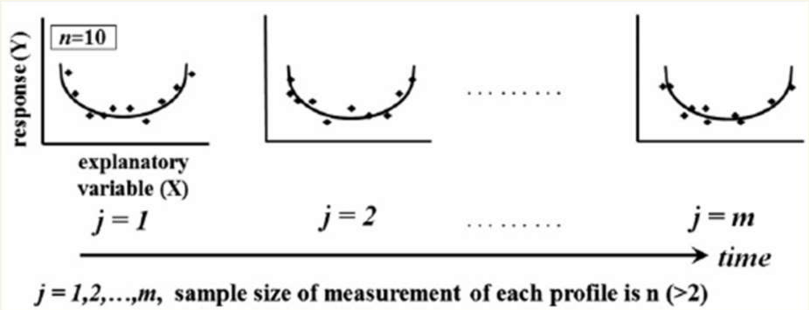
$$\boldsymbol{W}_j = \lambda\boldsymbol{Z}_j + (1-\lambda)\boldsymbol{W}_{j-1}, j = 1,2, \dots$$

  - **Upper Control Limits**

$$\boldsymbol{U}_j = \boldsymbol{W}_j^T\boldsymbol{\Sigma}^{-1}\boldsymbol{W}_j > L\frac{\lambda}{2-\lambda} \qquad \Sigma = \begin{bmatrix} (X^TX)^{-1} & 0 \\ 0 & 1 \end{bmatrix}$$

Non-Liner Profile Data

- The quality of a process or a product characterized by a functional relationship between the response variable (y) and one or more explanatory variables (x) is usually referred to as a profile.



Li, Chung-I., Jeh-Nan Pan, and Chun-Han Liao. "Monitoring nonlinear profile data using support vector regression method." *Quality and Reliability Engineering International* 35.1 (2019): 127-135.

Non-Liner Profile Data

- The underlying model

$$y_{ij} = f\left(x_{ij}\right) + \varepsilon_{ij}, i = 1,2, \dots, n_j, j = 1,2, \dots,$$

where $f(x)$ is a know function and the random errors ε_{ij} are assumed to follow some distributions.

- The following metrics are considered

$$M_{j1} = sign\left(\max_{i=1,\dots,n}\{\left|\hat{y}_{ij} - \tilde{y}_i\right|\}\right) \quad M_{j2} = \sum_{i=1}^{n}\left|\hat{y}_{ij} - \tilde{y}_i\right| \quad M_{j3} = \frac{1}{n}\sum_{i=1}^{n}\left|\hat{y}_{ij} - \tilde{y}_i\right| \quad M_{j4} = \left|M_{j1}\right| \quad M_{j5} = \sum_{i=1}^{n}\left(\hat{y}_{ij} - \tilde{y}_i\right)^2$$

where

1. The **predicted values** $\hat{y}_{ij}$ of the measurement y_{ij} for the corresponding explanatory variables are calculated based on the **SVR** model.
2. $\tilde{y}_i$ is the **reference profile** for the ith observation.

Non-Liner Profile Data

- Constructing the reference profile in Phase I study
  - A linear mode is considered

$$g(x,w) = w^T\phi(x) + b$$

where w is normal vector, $\phi(\boldsymbol{x})$ is a nonlinear transformation function and **b** is bias term

  - Suppose that we have the Phase I data $\{y_{ij}, i = 1, \dots, n, j = 1 \dots, g\}$
  - Support vector regression (SVR) model

$$min \frac{1}{2}\|w\|^2 + C\sum_{i=1}^{n}(\xi_i + \xi_i^*)$$

$subject\ to$:

$$\begin{cases} y_{ij} - g(x_i, w) \le \epsilon + \xi_i \\ g(x_i, w) - y_{ij} \le \epsilon + \xi_i^* \\ \xi_i, \xi_i^* \ge 0 \end{cases}$$

Non-Liner Profile Data

- Let $M_{j,p}, j = 1 \ldots,$ be the pth metric for the jth profile.
- The rank of M_j can be calculated as

$$R^*_{j,p} = 1 + \sum_{k=1}^{g-1} I\left(M_{j,p} > M^*_{k,p}\right)$$

where $\{M^*_{1,p}, M^*_{2,p}, \ldots, M^*_{g-1,p}\}$ are the reference metric and g-1 is the sample size of reference metric.

- The standardized rank of $\mathrm{R}^*_{\mathrm{j,p}}$ can be calculated as

$$R_{j,p} = \frac{2}{g}\left(R^*_{j,p} - \frac{g+1}{2}\right)$$

Non-Liner Profile Data

- Monitoring Statistics

$$EWMA_{j,p} = (1-\lambda)EWMA_{j-1,p} + \lambda R_{j,p} \ , j = 1,2, \ldots ,$$

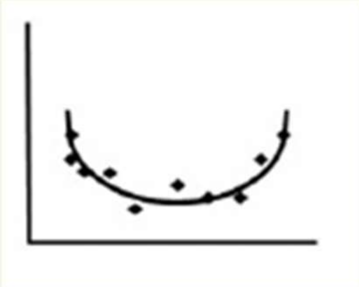
where $R_{j,p} = \frac{2}{g}\left(R^*_{j,p} - \frac{g+1}{2}\right)$ and $0 < \lambda < 1$ is a smoothing parameter

- The control limits are

$$\pm L_R \sqrt{\frac{\lambda g^2 - 1}{2 - \lambda 3 g^2}}$$

where L_R is determined based on Monte Carlo simulation to achieve a specified in-control average of run length.

Non-Liner Profile Data



- The reference profile

$$\tilde{y}_i = \sum_{j=1}^{g-1} \dot{y}_{ij}/(g-1)$$

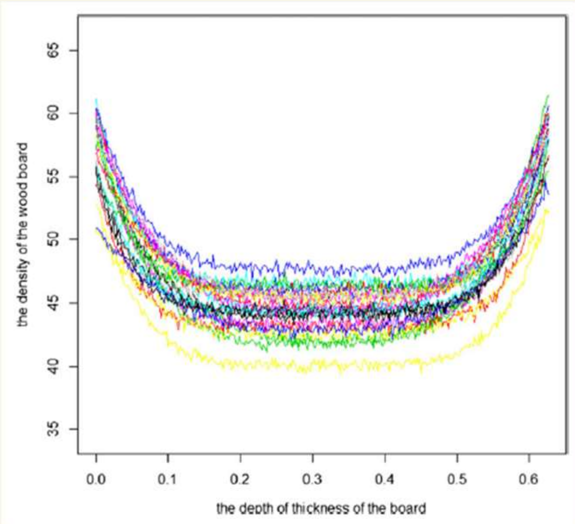
where $\dot{y}_{ij}$ is the predicted value for the observation y_{ij} that we have the Phase I data $\{y_{ij}, i = 1, \ldots, n, j = 1 \ldots, g\}$

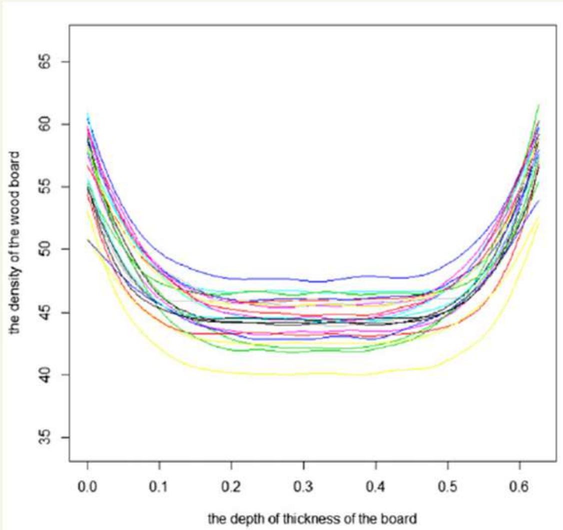
- The reference metric is

$$M_{j,2}^{*} = \sum_{i=1}^{n} \left|\dot{y}_{ij} - \tilde{y}_i\right|, j = 1, \ldots, (g-1)$$

Non-Liner Profile Data

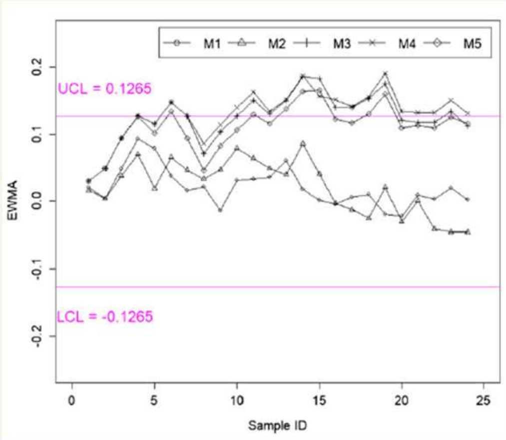
- A numerical example profile (vertical density profile, VDP)





Non-Liner Profile Data

- The nonparametric EWMA control chart

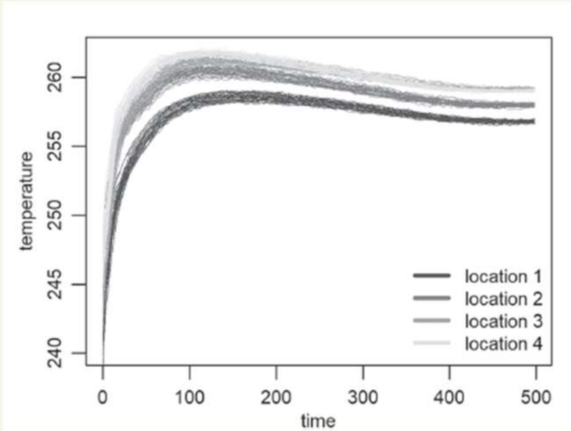


- Conclusion
  - The nonparametric EWMA control chart with M_1 and M_5 has better detecting performances

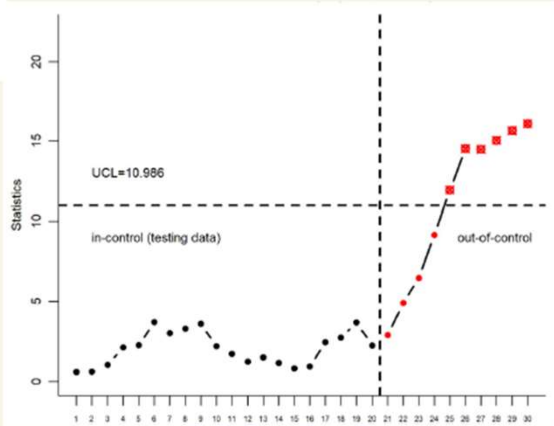
$$M_{j2} = \sum_{i=1}^{n} \left| \hat{y}_{ij} - \tilde{y}_i \right| \qquad M_{j5} = \sum_{i=1}^{n} \left(\hat{y}_{ij} - \tilde{y}_i \right)^2$$

Multivariate Non-Liner Profile Data

- A numerical example



- A nonparametric revised spatial rank EWMA control chart for phase II study



Non-Liner Profile Data

- The underlying model

$$\boldsymbol{Y}_j = \boldsymbol{F}_j + \boldsymbol{E}_j; j = 1,2,\dots,$$

- The data structure

$$\begin{bmatrix} y_{1j1} & \cdots & y_{1jp} \\ \vdots & \ddots & \vdots \\ y_{n_jj1} & \cdots & y_{n_jjp} \end{bmatrix}_{n_j \times p} = \begin{bmatrix} f_{1(x_{1j})} & \cdots & f_{p(x_{1j})} \\ \vdots & \ddots & \vdots \\ f_{1(x_{n_jj})} & \cdots & f_{p(x_{n_jj})} \end{bmatrix}_{n_j \times p} + \begin{bmatrix} \varepsilon_{1j1} & \cdots & \varepsilon_{1j} \\ \vdots & \ddots & \vdots \\ \varepsilon_{n_jj1} & \cdots & \varepsilon_{n_jjp} \end{bmatrix}_{n_j \times p}$$

- SRV (Support Vector Regression) metrics

$$w_{jk3} = \sum_{i=1}^{n_j} \left(y_{ijk} - \tilde{y}_{ik}\right)^2$$

where y_{ijk} is the i*th* observation with k*th* quality characteristic in the jth profile and $\tilde{y}_{ijk}$ is the value of the reference profile

Non-Liner Profile Data

- The vector of metrics

$$\boldsymbol{w}_{jd} = \left(\boldsymbol{w}_{j1d}, \boldsymbol{w}_{j2d}, \dots, \boldsymbol{w}_{jpd}\right)^T; j = 1,2, \dots,$$

- The spatial rank of $\boldsymbol{w}_j$

$$R_E\left(\widehat{\boldsymbol{M}}_{j-1}\boldsymbol{w}_{jd}\right) = \frac{1}{m_o + j - 1}\sum_{q=-m_0+1}^{j-1} U\left(\widehat{\boldsymbol{M}}_{j-1}\left(\boldsymbol{w}_{jd} - \boldsymbol{w}_{qd}\right)\right); \boldsymbol{w}_{(-m_0+j^*)d} = \boldsymbol{w}^*_{j^*d}, j^* = 1,2, \dots, m_0$$

  - $\hat{S}_{j-1}$ *is* the sample covariance matrix of the historical sample at the jth time point
  - $\widehat{M}_{j-1}$ is triangular Cholesky inverse root of $\hat{S}_{j-1}$

- The control statistic

$$Q_j^{R_E} = \frac{2-\lambda}{\lambda}\boldsymbol{v}_j^T\left\{\mathrm{cov}\left[R_F\left(\boldsymbol{M}\boldsymbol{w}_{jd}\right)\right]\right\}^{-1}\boldsymbol{v}_j$$

$$\mathrm{cov}\left[R_F\left(\boldsymbol{M}\boldsymbol{w}_{jd}\right)\right] = E\left[\left\|R_F\left(\boldsymbol{M}\boldsymbol{w}_{jd}\right)\right\|^2\right] I_{\mathrm{p}}/\mathrm{p}$$

$$\boldsymbol{v}_j = (1-\lambda)\boldsymbol{v}_{j-1} + \lambda R_E\left(\widehat{\boldsymbol{M}}_{j-1}\boldsymbol{w}_{jd}\right); \boldsymbol{v}_0 = \boldsymbol{0}$$

Non-Liner Profile Data

- The control statistic

$$Q_j^{R_E} = \frac{2-\lambda}{\lambda} \boldsymbol{v}_j^T \{\mathrm{cov}[R_F(\boldsymbol{M}\boldsymbol{w}_{jd})]\}^{-1} \boldsymbol{v}_j$$

where

$$\mathrm{cov}[R_F(\boldsymbol{M}\boldsymbol{w}_{jd})] = E\left[\|R_F(\boldsymbol{M}\boldsymbol{w}_{jd})\|^2\right] I_\mathrm{p}/\mathrm{p}$$

$$\boldsymbol{v}_j = (1-\lambda)\boldsymbol{v}_{j-1} + \lambda R_E(\widehat{\boldsymbol{M}}_{j-1}\boldsymbol{w}_{jd}); \boldsymbol{v}_0 = \boldsymbol{0}$$

- If

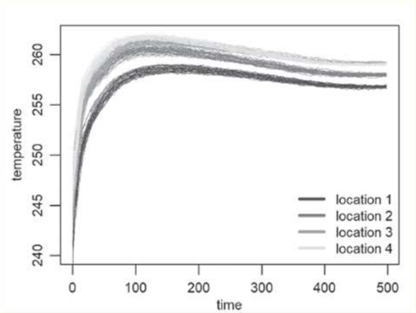
$$Q_j^{R_E} > L_{p,\lambda},$$

then the control chart will trigger an alarm. Otherwise, $\boldsymbol{w}_j$ will be included in the historical sample.

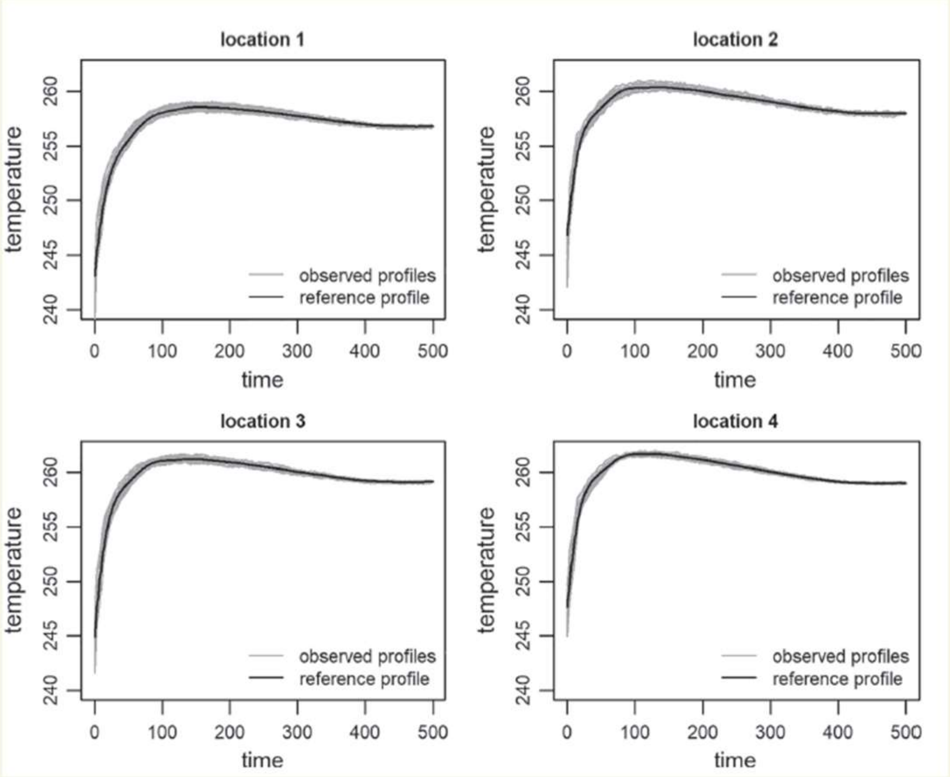
- The value of $L_{p,\lambda}$ needs to be determined to achieve a specified average run length.

Multivariate Non-Liner Profile Data

- A numerical example



- The reference profiles



Detecting the process changes for Non-Liner Profile Data

- Steps
  1. Define reference profile, $\tilde{y}$
  2. Calculate the predicted profile, $\hat{y}$
  3. Calculate the residuals, $\mathrm{y}^* = \tilde{y} - \hat{y}$
  4. Construct the control chart based on y^*

AI Algorithm